

A calibrated ATSC pilot-tone F-ratio detector and fixed-point GPU prototype for DTV flagging

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Abstract

Digital television (DTV) transmissions are persistent structured contaminants in the 400–800 MHz band used by several low-frequency radio astronomy experiments. ATSC 1.0 broadcasts provide a useful feature for targeted RFI flagging because each 6 MHz channel contains a narrow pilot near the lower channel edge. We present a scoped prototype study of a calibrated ATSC pilot-tone F-ratio detector. The implemented statistic uses three complex weight rows—one target row and two guard-separated reference rows—and forms $T = 2P_T/(P_{R-} + P_{R+})$, giving a constant-false-alarm-rate threshold axis under an effective F-distributed null model. The current implementation uses $K = 128$ fine bins, selected from a ~ 2 kHz pilot-frequency-offset design envelope and a power-of-two implementation constraint, $M = 1,048,576$ accumulated dot-product outputs set by the CHIME ring-buffer geometry, nominal ± 4 -bin references, and effective null degrees of freedom $\nu_1 = 2,097,152$ and $\nu_2 = 4,194,304$. We validate the null distribution with 300,000 simulated null trials, characterize injection sensitivity as a function of DTV data-shelf SNR, quantify frequency-offset and leakage robustness, and benchmark a fixed-point CUDA prototype. At the representative operating point $P_{FA} = 10^{-3}$, the threshold is $T = 1.003712$ and the empirical injection grid gives a required data-shelf SNR of -21.699 dB/bin for $P_D \simeq 0.99865$. The prototype GPU kernel operates directly on packed 4-bit real plus 4-bit imaginary CHIME-like samples, a format not served by standard cuBLAS complex dot-product interfaces, and reaches 216.23 blocks/s, corresponding to 29.02 GB/s of packed input payload or 232.18 Gb/s of raw input-bit throughput, but remains about a factor of 14.1 below the present real-time block-rate requirement. A 0.404 s CHIME baseband-derived segment demonstrates pilot-score generation, pilot-level flagging, expanded DTV-channel mask construction, threshold sweeps, and per-pilot pilot-power distributions over the strict nominal guard=4 publication subset. A +1.5 MHz shifted-grid field-background diagnostic on the same segment produces a much larger exceedance rate than the ideal analytic null, showing that the recorded segment contains substantial nearby spectral structure. This paper is intentionally scoped as a calibrated detector and implementation prototype; archive-scale validation, matched energy-detector comparison, clean field-null validation, and science-loss threshold selection are identified as required follow-on work rather than claimed results.

Keywords: radio-frequency interference; digital television; ATSC 1.0; radio astronomy instrumentation; constant false alarm rate; GPU implementation; 21 cm intensity mapping

1 Introduction

Radio-frequency interference (RFI) mitigation is a central requirement for modern radio astronomy. Human-made transmitters can raise the effective system temperature, bias power-spectrum measurements, contaminate time-domain searches, and force the excision of otherwise useful data. Common approaches include power thresholding, spectral-kurtosis statistics, generalized spectral-kurtosis tests, morphological flagging, reference-antenna cancellation, spatial filtering, and data-driven classifiers [1–6]. These methods are broadly useful, but many are intentionally agnostic to the spectral structure of a particular engineered signal.

This paper focuses on DTV interference from ATSC 1.0 broadcasts. DTV is persistent, geographically structured, and organized into 6 MHz channels. Unlike impulsive RFI, a DTV transmitter can remain continuously present while varying in received strength due to propagation, weather, and sidelobe coupling. Unlike an isolated narrowband contaminant, a DTV channel also contains a broad data shelf. The ATSC RF specification, however, includes a pilot at the suppressed-carrier frequency near the lower edge of each channel [7]. This known feature motivates a detector that identifies DTV using a pilot feature rather than total channel energy alone.

The motivating observational context is low-frequency radio astronomy in the 400–800 MHz band, including the Canadian Hydrogen Intensity Mapping Experiment (CHIME), which was designed for 21 cm intensity mapping and transient science [8]. In the CHIME cosmology context, baryon acoustic oscillation (BAO) measurements are sensitive to excess noise, residual foreground structure, and structured RFI leakage. The detector developed here is therefore intended to provide a calibrated RFI-feature measurement that can later be mapped to a science-loss budget.

We adopt the scoped prototype-paper framing. The goal is to present the detector, calibration, robustness diagnostics, fixed-point implementation, and a short recorded-data demonstration. Archive-scale telescope validation, comparison with a matched finite-sample energy detector, and science-optimized threshold selection are reserved for follow-on analyses requiring longer recorded-data coverage and a separate science-loss calculation. We include a compact shifted-grid recorded-data diagnostic, but we treat it as a field-background check rather than as archive-scale false-alarm validation.

2 Recorded DTV context and data provenance

Figure 1 introduces the signal context. Panel A is a recorded channelized spectrogram from a DTV-contaminated segment. Panel B compresses the same segment to a time-averaged spectrum and marks prominent narrowband or channel-associated features. Panel C gives an idealized ATSC 1.0 channel schematic, with the pilot near 0.309 MHz from the lower channel edge and the data shelf occupying most of the channel.

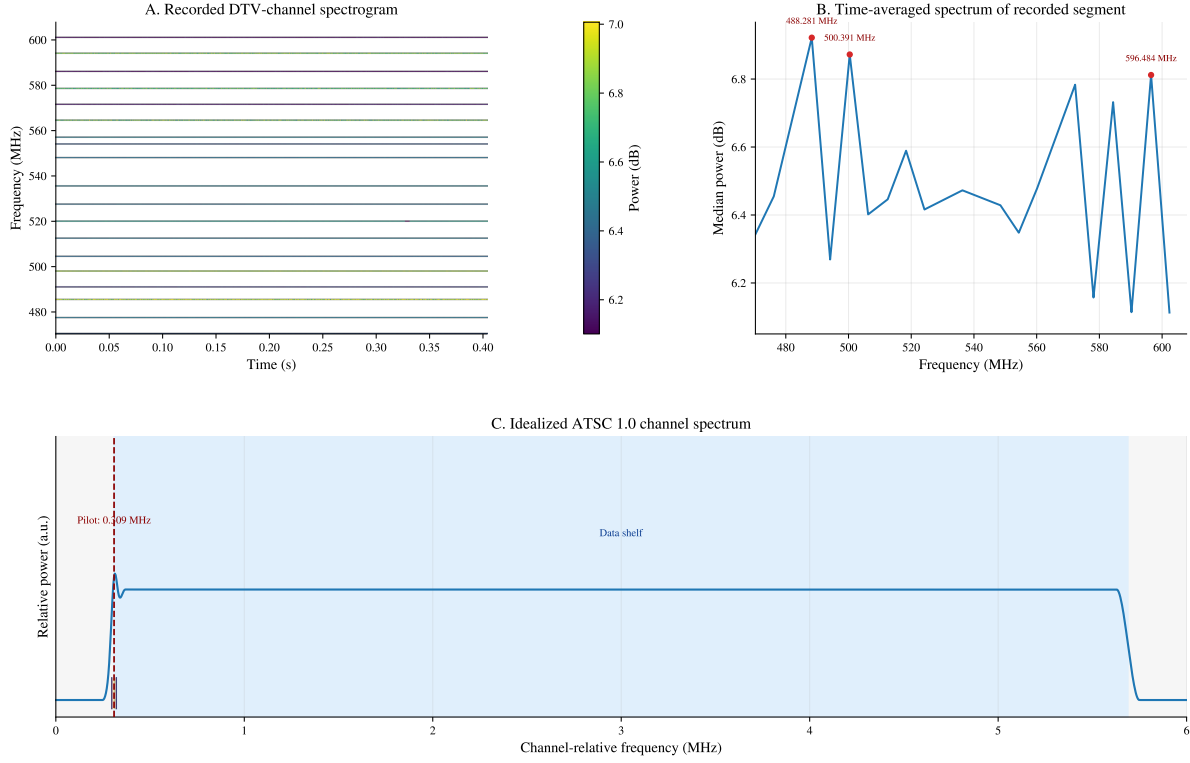


Figure 1: DTV signal context and ATSC 1.0 pilot structure. (A) Recorded channelized spectrogram from the demonstration segment. (B) Time-averaged spectrum of the same segment, with prominent narrowband or channel-associated features annotated. (C) Idealized ATSC 1.0 6 MHz channel schematic. The detector implementation uses a nominal pilot offset of 0.309441 MHz for mask expansion. Panels A and B use a recorded CHIME baseband-derived segment; Panel C is a generated schematic.

The recorded-data demonstration uses CHIME baseband-derived data accessed through the CANFAR Science Portal [9]. The data were processed in the `baseband-analysis:v1.9.0` container and drawn from the `chime-frb-public` project. Although the detector is motivated by CHIME cosmology, the demonstration uses CHIME/FRB baseband products because they provide minimally integrated data suitable for high-resolution RFI feature tests. The CHIME/FRB full-array baseband analysis pipeline is described by Michilli et al. [10].

The current recorded segment is approximately 0.404 s long, covers 470.312–607.812 MHz, and contains 4930 time blocks and 275 frequency channels in the offline spectrogram cache. The HDF5-to-NPZ cache is an offline plotting product, not the intended production interface. In a telescope deployment, the detector should operate on each input stream independently and emit stream-level pilot scores and masks. Aggregation across streams, polarizations, beams, and time windows should be treated as a downstream quality-assurance and science-policy step.

3 Detector statistic and mask construction

3.1 Signal and SNR convention

For an expected pilot frequency f_p , let $x_{p,j,m}$ denote the complex fine-channel sample for relative fine-bin index j and accumulation index m . The detector is designed around the local model

$$x_{p,j,m} = n_{p,j,m} + d_{p,j,m} + a_p h_j(\delta_p) e^{i\phi_m}, \quad (1)$$

where n is receiver noise, d is a noise-like DTV data-shelf component in the local channel, a_p is the pilot amplitude, $h_j(\delta_p)$ is the fine-bin response at fractional pilot offset δ_p , and ϕ_m is the pilot phase. The detector uses accumulated power only; no coherent pilot phase estimate is used in the statistic.

The injection results report performance as a function of *DTV data-shelf SNR*, defined as the per-fine-bin shelf-to-noise power ratio

$$\text{SNR}_{\text{shelf}} = 10 \log_{10} (P_{\text{shelf}}/P_N) \quad \text{dB/bin}. \quad (2)$$

The controlled injection convention converts this shelf SNR to pilot-bin SNR using the metadata in Table 1. In particular, the injected pilot is 11.7 dB below the integrated DTV data power, while the shelf is represented over 16 fine bins. Therefore the pilot power is $10 \log_{10}(16) - 11.7 = 0.3412$ dB above one shelf fine bin before frequency-response loss. Thus, for a centered pilot,

$$\text{PNR}_{\text{pilot}} = \text{SNR}_{\text{shelf}} + 0.3412 \text{ dB} \quad (3)$$

under the injection model used for Fig. 4. This convention is the abscissa for the sensitivity figures; it is not a measurement of absolute received DTV-channel power in the short CHIME segment.

Table 1: Signal-normalization convention used for the injection sensitivity curves.

Quantity	Value or definition
Noise power per fine bin	P_N
DTV shelf SNR	$10 \log_{10}(P_{\text{shelf}}/P_N)$ dB/bin
DTV shelf support	16 fine bins
Pilot offset from lower ATSC edge	0.309441 MHz
Pilot-to-integrated-data power	-11.7 dB
Pilot-to-shelf-per-bin power	+0.3412 dB
Centered-pilot PNR	$\text{SNR}_{\text{shelf}} + 0.3412$ dB
Frequency-response factor	$ h_j(\delta_p) ^2$, evaluated by the fine-bin response model
Detector phase use	accumulated-power only; no coherent phase estimator

3.2 F-ratio statistic

For pilot p , the implemented statistic uses three complex weight vectors: a target row $w_{p,k}^{(T)}$ and two reference rows $w_{p,k}^{(R-)}$ and $w_{p,k}^{(R+)}$. These rows select the pilot-centered target response and the two guard-separated references. The row outputs are

$$z_{p,m}^{(r)} = \sum_{k=0}^{K-1} x_{p,k,m} \left(w_{p,k}^{(r)} \right)^*, \quad r \in \{T, R-, R+\}. \quad (4)$$

The corresponding accumulated powers are

$$P_p^{(r)} = \sum_{m=1}^M \left| z_{p,m}^{(r)} \right|^2. \quad (5)$$

The F-ratio statistic is

$$T_p = \frac{2P_p^{(T)}}{P_p^{(R-)} + P_p^{(R+)}}. \quad (6)$$

For the nominal geometry, the target row is centered on the expected pilot bin and the reference rows are located at relative offsets -4 and $+4$ fine bins. The detector configuration therefore has $N_{\text{weight}} = 3$ rows, not three independent target bins.

Under the current null calibration,

$$T_p \mid H_0 \sim F(\nu_1, \nu_2), \quad (7)$$

with $\nu_1 = 2M$ and $\nu_2 = 4M$. The constant-false-alarm-rate threshold is therefore

$$\eta(P_{\text{FA}}) = F_{\nu_1, \nu_2}^{-1}(1 - P_{\text{FA}}), \quad (8)$$

where P_{FA} is the one-sided survival probability under H_0 .

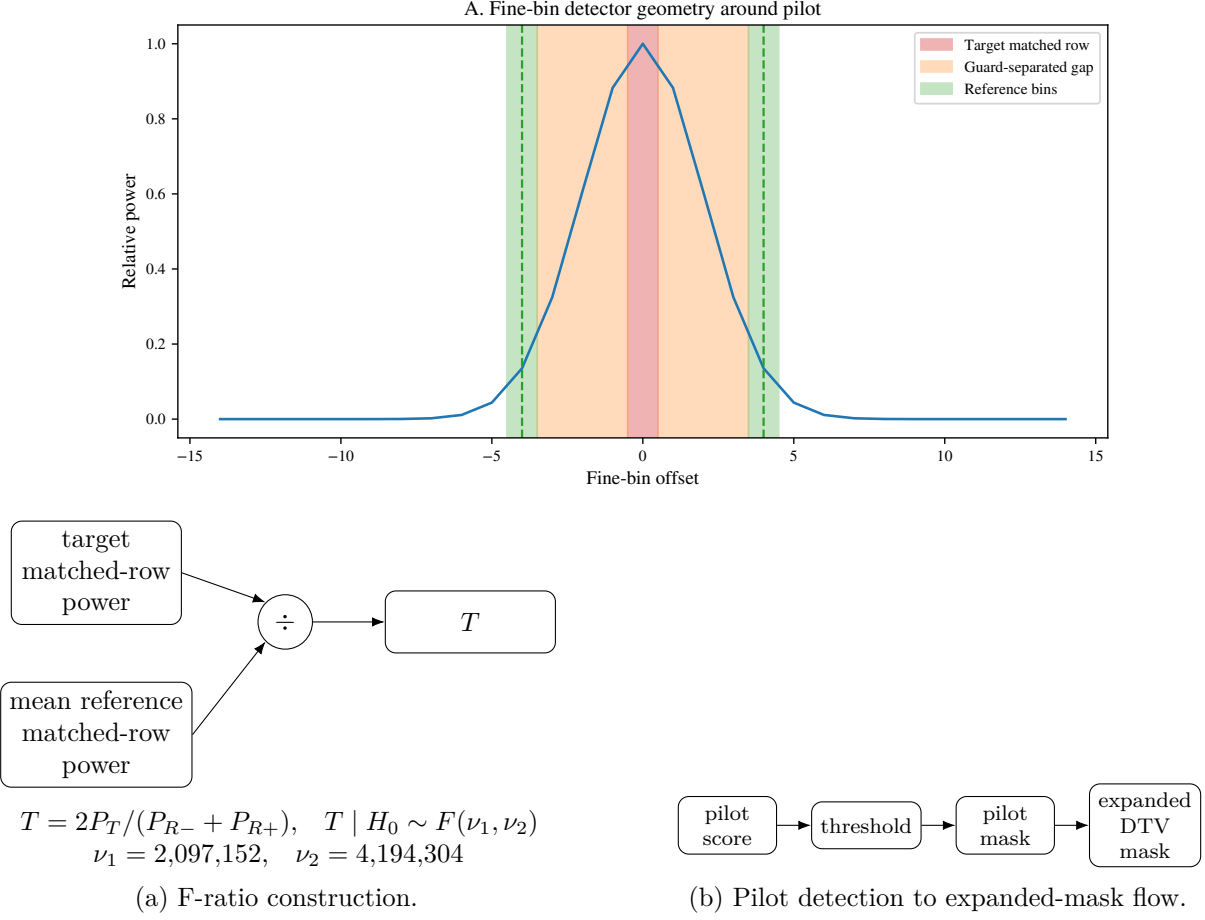


Figure 2: Detector geometry and score construction. The generated panel shows the fine-bin geometry around a pilot. The implemented statistic has one target matched row and two guard-separated reference rows, with nominal relative reference offsets $\{-4, +4\}$. Thresholding is performed on the linear F-ratio T ; score maps use the logarithmic display convention defined in Sec. 3.4.

3.3 Effective degrees of freedom and edge-channel references

The degrees of freedom in Eq. (7) follow from the row statistic that is actually implemented. A complex Gaussian matched-row output has two real degrees of freedom per accumulated sample. One target row accumulated over $M = 1,048,576$ samples gives $\nu_1 = 2M = 2,097,152$. Two reference rows give $\nu_2 = 4M = 4,194,304$. The calibration should therefore be interpreted as an effective-F model for one target row and two reference rows, not as a three-target-bin independent-bin average.

The nominal reference geometry is ± 4 fine bins, corresponding to $\pm 12,207.03$ Hz for $K = 128$. The synthetic null, sensitivity, and robustness diagnostics use this nominal geometry. The source weight manifest also identifies two boundary-safe edge cases in the available channel product: 512.309 MHz would use effective ± 1 -bin references and 578.309 MHz would use effective ± 3 -bin references. These rows are excluded from the nominal guard=4 recorded-data publication subset. Consequently, the recorded-data metrics and histogram products report 17 nominal guard=4 pilot rows, while the two non-nominal edge rows and four missing-channel rows are explicitly marked in the result metadata and pilot histogram.

3.4 Score-map display convention

All detector decisions are made on T using Eq. (8). Some figures display a logarithmic pilot-excess score,

$$S_{p,\text{dB}} = 10 \log_{10}[\max(T_p - 1, \epsilon)], \quad (9)$$

where ϵ is a display floor. The plotted score maps use $\epsilon = 10^{-5}$, equivalent to a -50 dB display floor. This display emphasizes excess target power above the local reference. It is not the same as $10 \log_{10} T_p$. At the default $P_{\text{FA}} = 10^{-3}$ operating point, $T = 1.003712$ corresponds to $10 \log_{10} T = 0.0161$ dB but to $10 \log_{10}(T - 1) = -24.30$ dB under the pilot-excess display convention. This distinction explains why pilot-score maps can have negative dB color bars even though thresholding is performed on T near unity.

3.5 Pilot-to-DTV mask expansion and false-alarm burden

Let $b_p(t) = \mathbf{1}\{T_p(t) > \eta\}$ be the pilot-level detection for expected pilot p at time block t . The DTV mask expansion used here adopts the ATSC channel width $W = 6.0$ MHz and implementation pilot offset $f_{\text{pilot}} = 0.309441$ MHz. The lower edge associated with pilot p is

$$f_{\ell,p} = f_p - f_{\text{pilot}}. \quad (10)$$

Ignoring optional padding, the expanded mask is

$$m(f, t) = \bigvee_p b_p(t) \mathbf{1}[f \in (f_{\ell,p}, f_{\ell,p} + W)], \quad (11)$$

with clipping to the available frequency range. The analysis uses an expansion mode labeled `dtv_channel_band`, channel width 6 MHz, pilot offset 0.309441 MHz, and nearest-channel mapping with a 0.75 MHz tolerance. In the plotted recorded-data range, all pilot ranges are mapped to the available channelized product.

A per-test false-alarm rate is not an operational false-mask rate. If N_p expected pilots, N_t time blocks, and N_s independent streams are tested, then the expected number of pilot-level null detections is

$$\mathbb{E}[N_{\text{false}}] = N_p N_t N_s P_{\text{FA}}. \quad (12)$$

In the single-stream recorded demonstration, the strict guard=4 publication subset has $N_p = 17$ nominal pilot channels and $N_t = 4930$ time blocks, so $P_{\text{FA}} = 10^{-3}$ would correspond to roughly 84 pilot-level null detections if all tests were true-null and independent. The recorded segment is not a true-null sample, but this calculation shows why telescope-scale aggregation needs persistence logic, cross-stream consistency checks, or science-driven threshold selection.

4 Selected detector configuration

The detector operating point is set by three instrument-facing constraints: the expected pilot-frequency-offset envelope, the amount of data available in the ring buffer, and the low-bit complex format of the CHIME voltage product. This section records those choices explicitly so that the numerical parameters in the calibration are not treated as arbitrary tuning constants.

4.1 Fine-bin factor from the frequency-offset envelope

The fine-bin parameter K sets detector frequency granularity and therefore the loss incurred when a received pilot is displaced from the fine-bin center. In the current data products, aircraft scatter and related propagation effects can produce pilot-frequency offsets of order 2 kHz. We therefore use a design frequency resolution of order 4 kHz, so that a worst-case 2 kHz offset is comparable to a half-bin displacement rather than a multi-bin displacement.

The relevant CHIME coarse-channel spacing for this detector is approximately

$$\Delta f_{\text{coarse}} = 390.625 \text{ kHz}. \quad (13)$$

A target fine-bin width of $\Delta f_{\text{target}} \simeq 4 \text{ kHz}$ gives

$$K_{\text{ideal}} = \frac{\Delta f_{\text{coarse}}}{\Delta f_{\text{target}}} \simeq \frac{390.625}{4} \simeq 97.7. \quad (14)$$

The implementation selects the nearest convenient power-of-two value above this design point,

$$K = 128, \quad (15)$$

giving

$$\Delta f_K = \frac{390.625 \text{ kHz}}{128} = 3051.76 \text{ Hz} \quad (16)$$

and a half-bin scale of 1525.88 Hz. Thus the 2 kHz design envelope is not assumed to be loss-free; the corresponding offset loss is measured explicitly in the robustness diagnostics. This choice trades frequency localization against Doppler-like offset tolerance and compute cost. Smaller K would provide broader bins but coarser localization, while larger K would provide finer localization at higher computational cost and with greater sensitivity to pilot-frequency mismatch.

4.2 Accumulation length from the ring buffer

The accumulated sample count M is also instrument-derived. Each antenna-polarization stream provides 2^{16} complex samples in the ring-buffer product used for the prototype calculation. With $K = 128 = 2^7$, each stream is logically divided into

$$N_{\text{subblock}} = \frac{2^{16}}{128} = 512 \quad (17)$$

subblocks for the complex dot product. Accumulating those dot-product powers over 1024 antennas and two polarizations gives

$$M = N_{\text{subblock}}N_{\text{ant}}N_{\text{pol}} = 512 \times 1024 \times 2 = 1,048,576. \quad (18)$$

This value sets the effective null degrees of freedom for the one-target/two-reference statistic: $\nu_1 = 2M$ and $\nu_2 = 4M$. In deployment, M is therefore a latency/sensitivity parameter: longer accumulation tightens the null distribution and improves weak-pilot sensitivity, while shorter accumulation reduces latency but broadens the field distribution.

4.3 Guard spacing and current publication operating point

The nominal reference rows are placed at $\{-4, +4\}$ fine bins relative to the target row, corresponding to a 12.207 kHz offset on either side for $K = 128$. The guard was selected as the minimum guard satisfying the current leakage criterion in the synthetic robustness sweep. A guard that is too small is dangerous because pilot leakage can contaminate the reference denominator and suppress the detector statistic. A guard that is too large usually has a smaller leakage penalty, but it can become less local and more sensitive to spectral curvature, bandpass structure, and edge-channel availability. For this reason we retain guard=4 for the present calibrated operating point and list a wider guard sweep as follow-on validation rather than silently changing the detector.

Table 2 summarizes the operating point used throughout the paper.

Table 2: Selected detector parameters for the scoped prototype paper.

Quantity	Value
Coarse-channel spacing used by detector	390.625 kHz
Design pilot-frequency-offset envelope	~ 2 kHz
Target fine-bin width before radix-2 rounding	~ 4 kHz
Fine-bin parameter	$K = 128$
Fine-bin width	3051.76 Hz
Half-bin frequency scale	1525.88 Hz
Ring-buffer samples per antenna-polarization stream	2^{16}
Antenna streams and polarizations	1024 antennas, 2 polarizations
Dot-product subblocks per stream	512
Accumulated dot-product outputs	$M = 1,048,576$
Input quantization metadata	packed 4-bit real + 4-bit imaginary complex samples
Weight-vector count	3: target, lower reference, upper reference
Implemented statistic	$T = 2P_T/(P_{R-} + P_{R+})$
Nominal reference locations	$\{-4, +4\}$ fine bins
Nominal guard/reference spacing	four fine bins; 12,207.03 Hz
Pilot-to-shelf-per-bin injection offset	+0.3412 dB
Effective null degrees of freedom	$\nu_1 = 2,097,152$, $\nu_2 = 4,194,304$
Default operating point	$P_{\text{FA}} = 10^{-3}$, $\eta = 1.003712$
Threshold in $10 \log_{10} T$	0.0161 dB
Threshold in $10 \log_{10}(T - 1)$	-24.30 dB
Score-map display floor	-50 dB

5 Null calibration and CFAR thresholding

The analytic null model is useful only if it matches the implemented statistic at the tail depths relevant to mask generation. Figure 3 compares the simulated null distribution with the effective F model. The null run used 300,000 trials. The Kolmogorov–Smirnov statistic is 0.00126 with $p = 0.727$, indicating no significant discrepancy at the sampled resolution. With a 10-tail-sample rule, the simulation resolves survival probabilities down to about 3.3×10^{-5} ; 10^{-4} remains empirically sampled with roughly 30 expected tail samples, while lower P_{FA} values are increasingly theory-supported rather than directly resolved.

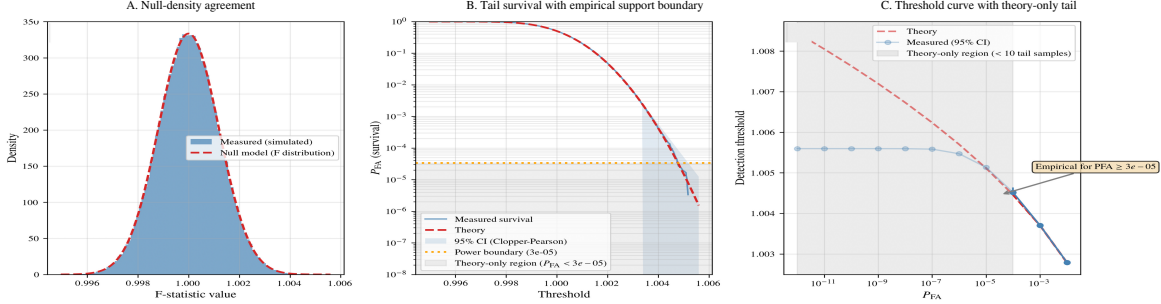


Figure 3: Null-distribution and CFAR calibration for the $K = 128$ detector. (A) Simulated null-density agreement with the effective F-distribution model. (B) Tail survival with empirical support boundary. (C) Detection threshold as a function of target false-alarm probability. The simulated null run used 300,000 trials; low- P_{FA} thresholds beyond the empirically sampled tail should be treated as theory-supported.

We use $P_{\text{FA}} = 10^{-3}$ as a representative engineering operating point for detector characterization. This is an engineering operating point. For BAO applications, the threshold should be selected by propagating mask burden and residual contamination into a science-loss metric rather than by choosing a convenient detector false-alarm rate.

6 Injection sensitivity

Figure 4 summarizes sensitivity over the current injection grid. The heat map uses 12,000 trials on the displayed SNR and false-alarm grid. At $P_{\text{FA}} = 10^{-3}$, the empirical data-shelf SNR required for the $P_{\text{D}} = 0.9986501$ target is -21.699 dB/bin. The corresponding estimate has 11,983 detections in 12,000 trials and a 95% Clopper–Pearson interval of 0.9977–0.9992. Higher- P_{D} curves are labeled by their detection probabilities rather than by Gaussian “sigma” shorthand because the finite grid saturates when the expected number of misses becomes less than one.

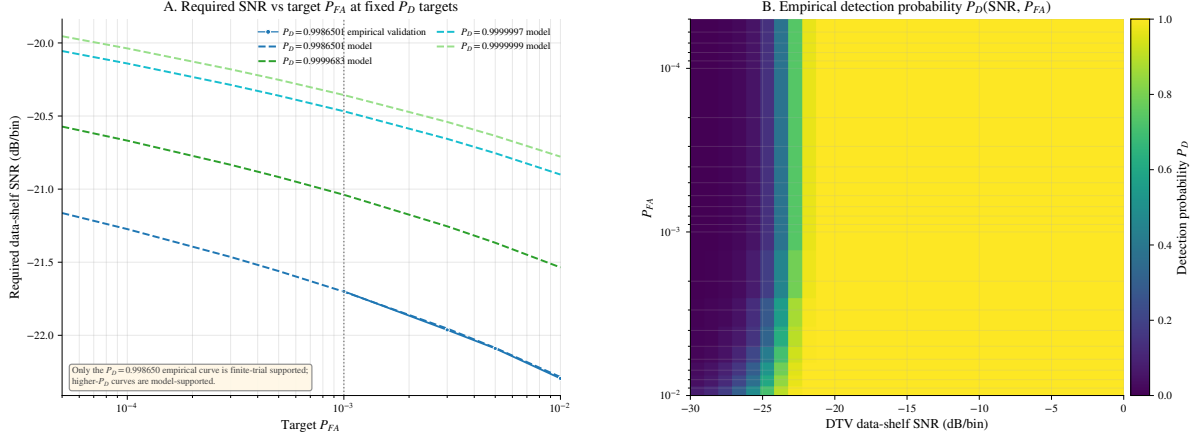


Figure 4: Detection sensitivity for the $K = 128$, guard=4 detector. (A) Required DTV data-shelf SNR versus target false-alarm probability for several target detection probabilities. The $P_D = 0.998650$ empirical curve is directly supported by finite trials. The $P_D = 0.999968$, 0.9999997 , and 0.9999999 curves are model-supported because the current 12,000-trial injection grid has insufficient miss statistics to resolve probabilities this close to unity. (B) Empirical detection probability versus data-shelf SNR and P_{FA} .

The sensitivity result should be interpreted as detector-side performance under the injection convention in Table 1. A matched local-energy baseline and a BAO-tolerable residual RFI level are reserved for the deployment-validation analysis. Those comparisons are central follow-on analyses for a deployment-validation paper.

7 Robustness diagnostics

Figure 5 tests departures from the idealized centered-pilot model. Panel A shows the rectangular-bin sinc loss for within-bin offsets. The frequency-offset diagnostic in Panel B measures pilot-frequency mismatch; terrestrial Doppler is only one possible contributor to such mismatch. Panel C shows that the measured false-alarm rate remains near 10^{-3} over the tested ± 3 dB gain/noise perturbation range, varying from about 8.1×10^{-4} to 1.2×10^{-3} . Panel D supports the selected four-bin guard under the current reference-leakage criterion. Because undersized guards can contaminate the reference while oversized guards can become less local, a future publication or deployment sweep should test wider guards such as $g = 6, 8, 10, 12$, and 16 against both pilot-leakage loss and recorded-data field-background behavior.

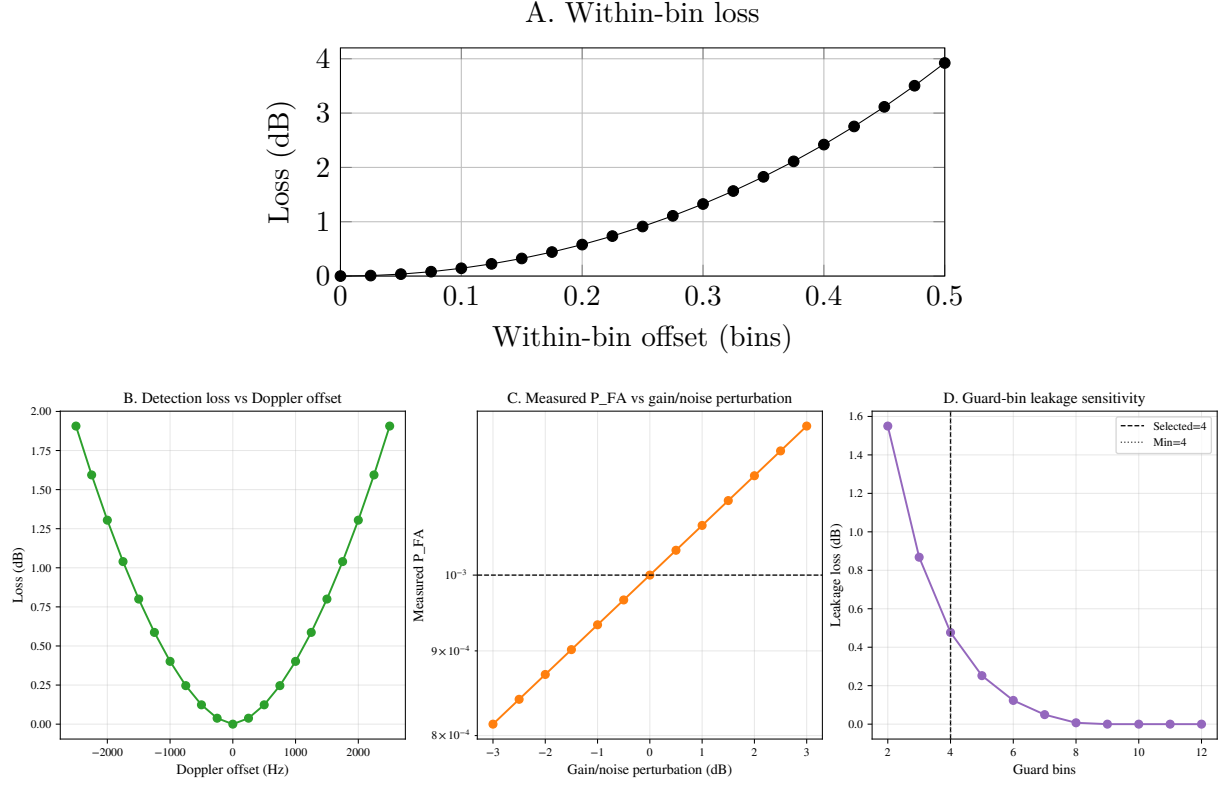


Figure 5: Robustness diagnostics for the selected $K = 128$, guard=4 detector. Panel A shows analytic within-bin sinc-response loss for a rectangular fine bin. Panels B–D summarize pilot-frequency-offset loss, measured false-alarm rate under gain/noise perturbation, and guard-bin leakage sensitivity. Panel D supports the selected four-bin guard under the current leakage criterion; wider-guard tests remain a useful deployment-design sweep.

The sinc-response model in Panel A is correct for an unwrapped rectangular fine-bin response. If a deployment uses a different fine-channelizer or a PFB-derived response at this stage, the offset-loss model should be replaced by the measured fine-channel response. The detector calibration itself is local and empirical; the plotted analytic curve is a response model, not an assumption required by the CFAR threshold.

8 Fixed-point CUDA prototype

A deployment-oriented detector must operate on high-rate, low-bit complex data. The implementation is not a direct call to a vendor BLAS routine: the CHIME-like voltage samples used here are represented as packed 4-bit real and 4-bit imaginary components, while the detector requires complex dot products with three complex weight rows per pilot. Standard cuBLAS dot-product and GEMV interfaces do not provide this packed complex 4+4-bit operation, so the prototype uses a custom CUDA kernel that unpacks samples, evaluates the target and reference projections, accumulates powers, and forms the F-ratio.

Figure 6 shows the instrument-to-kernel dataflow. The 2^{16} -sample ring-buffer product is split into $K = 128$ -sample logical subblocks for each antenna-polarization stream. For each subblock m and weight row n , the kernel computes

$$z_{m,n} = \sum_{k=0}^{K-1} x_{m,k} w_{n,k}^*, \quad (19)$$

where $n = 0$ is the target row and $n = 1, 2$ are the lower and upper reference rows. It then accumulates

$$P_n = \sum_{m=1}^M |z_{m,n}|^2 \quad (20)$$

and emits

$$T = \frac{2P_0}{P_1 + P_2}. \quad (21)$$

This is the same statistic used in the analytic calibration.

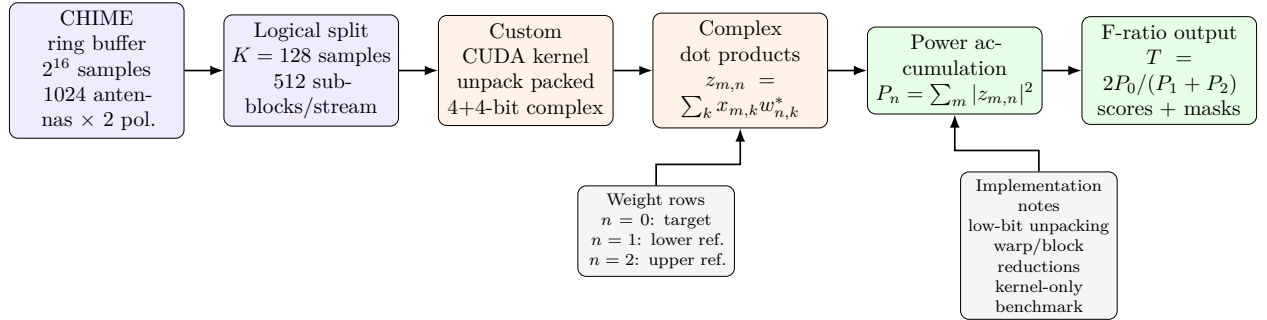


Figure 6: Ring-buffer-to-kernel dataflow for the prototype implementation. The detector operates on packed 4-bit real plus 4-bit imaginary complex samples. The ring-buffer product is logically split into $K = 128$ -sample subblocks for each antenna-polarization stream. The custom CUDA kernel evaluates target and reference complex dot products, accumulates row powers over the available subblocks, antennas, and polarizations, and emits the same F-ratio statistic used in the analytic detector calibration.

Figure 7 summarizes the current fixed-point prototype benchmark and numerical comparison. The benchmark is kernel-only and excludes host/device transfers. It is therefore an implementation characterization rather than a final deployment throughput measurement.

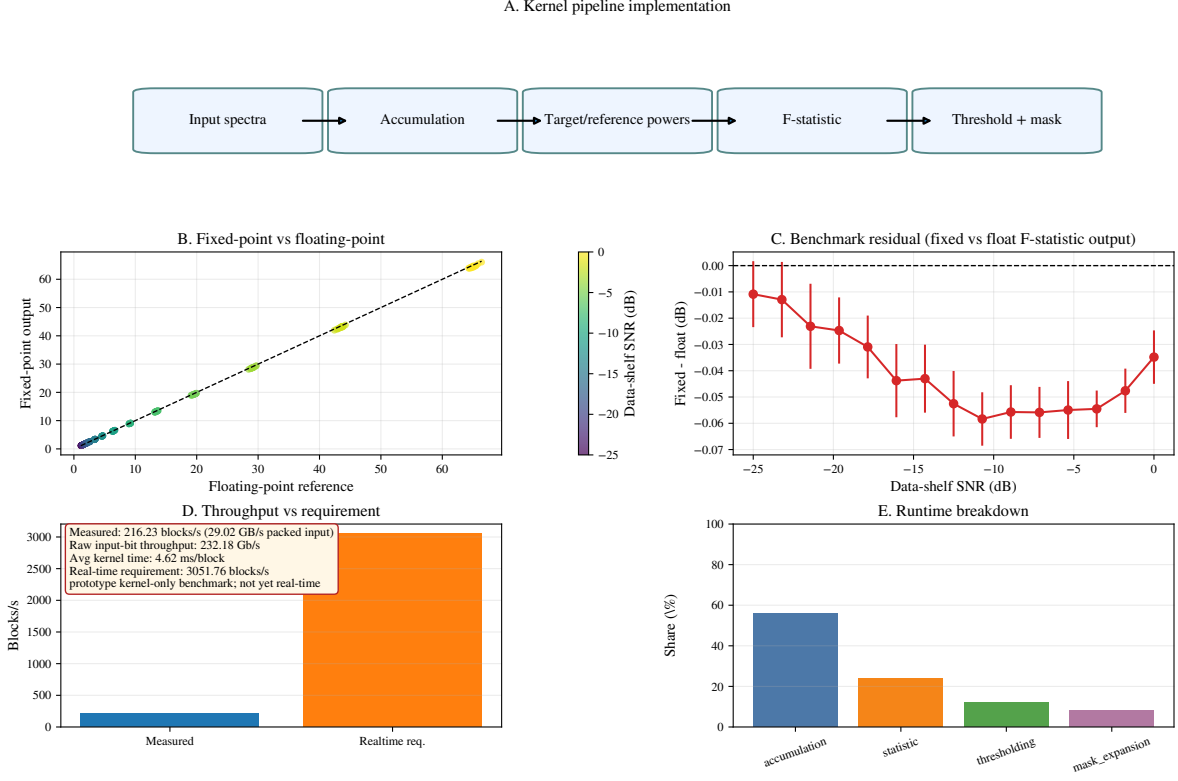


Figure 7: Prototype CUDA/fixed-point implementation characterization. Panel A shows the kernel processing flow. Panel B compares fixed-point output with the floating-point reference. Panel C gives the fixed-minus-float F-ratio residual over the tested SNR range. Panel D compares measured kernel-only throughput with the real-time block-rate requirement. Panel E gives the runtime breakdown. Host/device transfers are excluded.

The benchmark reports 216.23 blocks/s with an average kernel time of 4.62 ms/block. The packed input payload per block is MK one-byte complex samples, giving

$$N_{\text{payload}} = 1,048,576 \times 128 = 134,217,728 \text{ bytes.} \quad (22)$$

The resulting throughput is

$$R_{\text{byte}} = 216.23 \times 134,217,728 / 10^9 = 29.02 \text{ GB s}^{-1}, \quad (23)$$

or 232.18 Gb/s when expressed as raw input-bit throughput. The current real-time requirement is 3051.76 blocks/s, so the prototype is approximately a factor of 14.1 below the block-rate requirement.

The fixed-point comparison quantifies bounded implementation loss rather than exact floating-point equivalence. The direct F-statistic residual in Fig. 7 is small over the tested SNR range. A companion implementation-loss characterization reports a data-shelf-equivalent mean loss of approximately -0.309 dB and a maximum absolute loss of approximately 0.59 dB over the measured range. These values are tied to the tested bit depth, input scaling, and kernel implementation; they should not be treated as universal corrections outside the validated interval.

The current kernel is intentionally a prototype. The natural optimization path is to fuse multiple pilots per launch, stage the small weight table in constant or shared memory, reduce synchronization overhead with warp-level reductions, and integrate the detector with upstream data movement so that host/device transfer costs and stream scheduling are included in the end-to-end benchmark. Those steps are deployment-engineering work and are separated from the detector-calibration result in this paper.

9 Single-segment recorded-data demonstration

Figure 8 shows the detector applied to the recorded CHIME segment. Panel A shows the channelized spectrogram. Panel B shows the pilot-excess score map using the display convention in Eq. (9). Panel C shows the raw pilot F-statistic detection mask. Panel D shows the expanded DTV-channel mask. Panel E summarizes the threshold-exceedance fraction by pilot frequency, which is more informative for this nearly stationary short segment than a global masked-fraction time series.

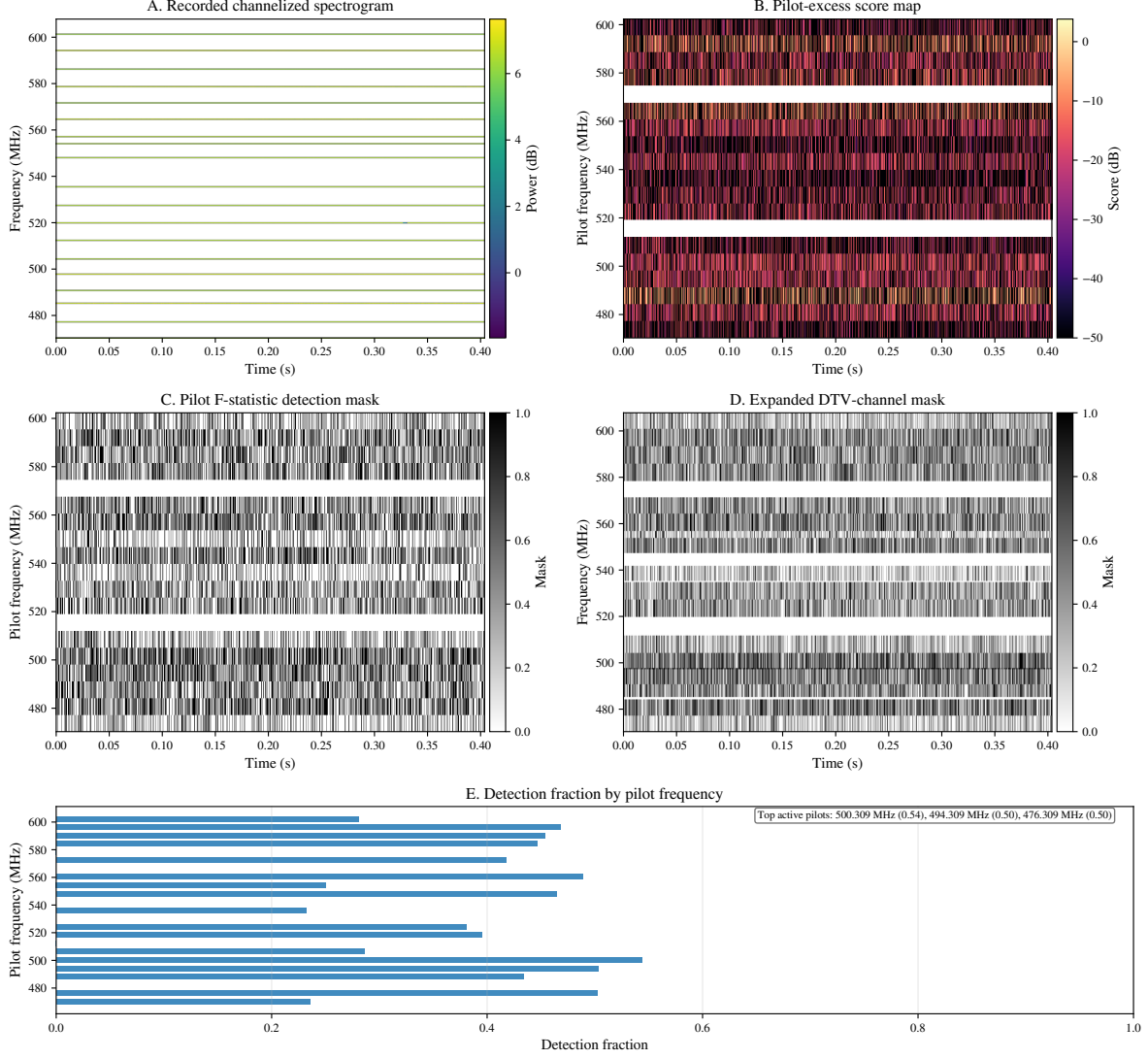


Figure 8: Single-segment recorded-data demonstration. (A) Recorded channelized spectrogram. (B) Pilot-excess score map using $10 \log_{10}[\max(T - 1, 10^{-5})]$. (C) Pilot F-statistic detection mask, defined as $T > \eta$. (D) Expanded DTV-channel mask using the 6 MHz channel-band expansion in Eq. (11). (E) Detection fraction by pilot frequency. The operating point is $P_{FA} = 10^{-3}$ with threshold $T = 1.003712$, $K = 128$, and nominal guard=4. The top active pilots in this segment are 500.309, 494.309, and 476.309 MHz, with detection fractions 0.54, 0.50, and 0.50. This figure demonstrates detector output on one recorded segment and is not archive-scale validation.

The figure demonstrates that the detector produces structured pilot-level and expanded masks on recorded data. At the default threshold, the all-pixel expanded-mask fraction in the plotted product is 0.3404, the mapped-DTV-bin masked fraction is 0.4052, and the valid-pixel unmasked fraction is 0.6430. These numbers are engineering diagnostics for the short segment, not long-term observatory mask-burden estimates. Long-term stability, seasonality, propagation dependence, and deployment performance require the larger validation run described below.

9.1 Per-pilot pilot-power distributions

Figure 9 summarizes the distribution of pilot-excess scores for each expected ATSC pilot. This figure complements the binary mask in Fig. 8: the mask reports threshold exceedance, whereas the histogram shows the amplitude distribution that produced those detections. The full expected pilot table contains 23 pilot frequencies from 470.309 to 602.309 MHz. In the current channelized cache, 17 pilots satisfy the nominal guard=4 publication geometry. Four expected pilots are missing from the cache: 482.309, 530.309, 542.309, and 566.309 MHz. Two additional pilots, 512.309 and 578.309 MHz, are excluded from the nominal subset because their source weight-table rows would require non-nominal edge-channel references. Missing and excluded rows are shown explicitly rather than silently omitted.

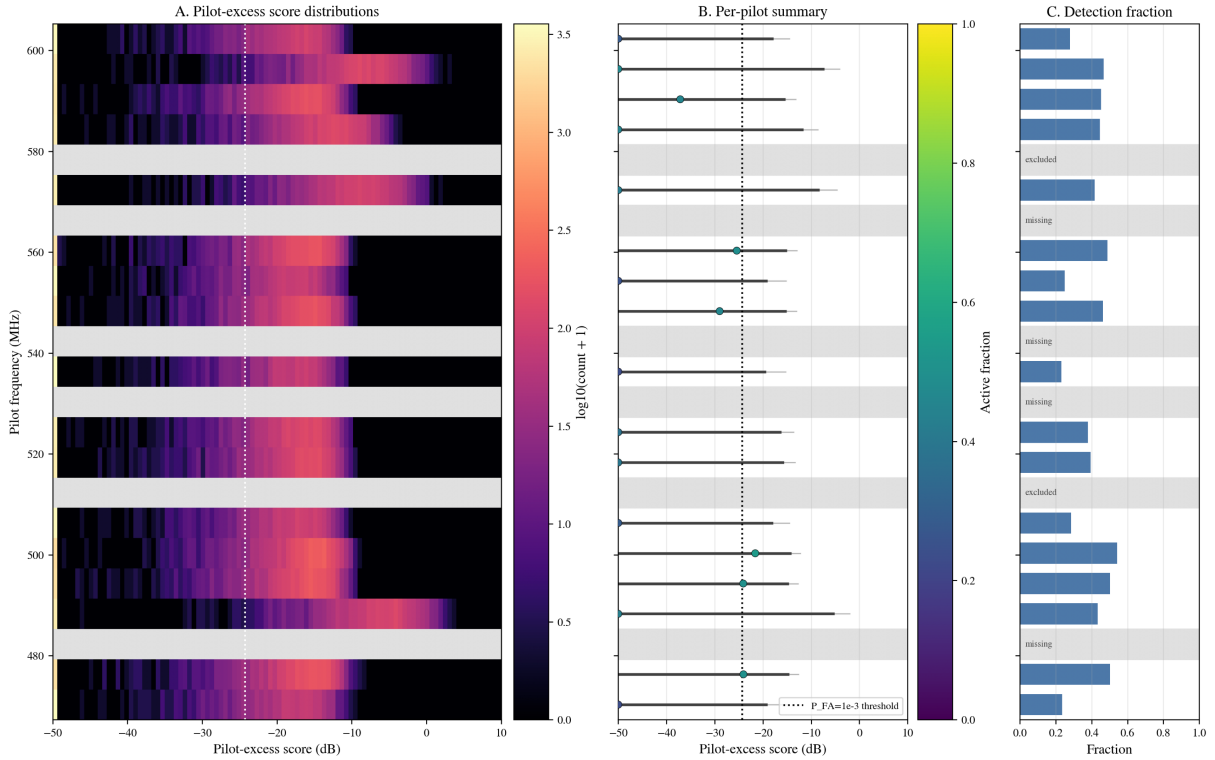


Figure 9: Per-pilot pilot-excess score distributions for the expected ATSC pilot grid. Each row in Panel A gives the histogram of $10 \log_{10}[\max(T - 1, 10^{-5})]$ for one expected pilot frequency. The vertical dotted line marks the $P_{FA} = 10^{-3}$ threshold, $10 \log_{10}(\eta - 1) = -24.30$ dB. Panel B summarizes per-pilot quantiles, and Panel C gives the raw threshold-exceedance fraction. Gray rows mark expected pilots whose corresponding channel product is unavailable in this cache or excluded from the nominal guard=4 publication subset.

The histogram shows that the strongest detections are not simply isolated outliers; several pilot frequencies have broad distributions with substantial probability mass to the right of the threshold. The largest exceedance fraction is at 500.309 MHz, followed by 494.309 and 476.309 MHz. Several rows also contain a large fraction at the -50 dB display floor, indicating blocks for which $T - 1 \leq 10^{-5}$ after clipping. The floor bin is kept in the main figure because it is part of the detector-output distribution.

9.2 Threshold sweep on the recorded segment

The threshold sweep in Fig. 10 shows how recorded-segment metrics change with the target false-alarm probability. The sweep is useful as an engineering diagnostic because it separates all-pixel mask burden, mapped-DTV-channel mask burden, flagged-bin retained fraction, suppression in the flagged-bin diagnostic, and active-pilot count. It is not a substitute for long-duration validation.

At the default $T = 1.003712$ threshold, the all-pixel expanded-mask fraction is 0.3404 and the mapped-DTV-bin masked fraction is 0.4052. The flagged-bin retained fraction is 0.6010 and the flagged-bin suppression diagnostic is 2.23 dB. The active-pilot count is 17 across the displayed threshold grid. These domain-explicit quantities are intentionally not forced to be complements: they are computed over different diagnostic domains.

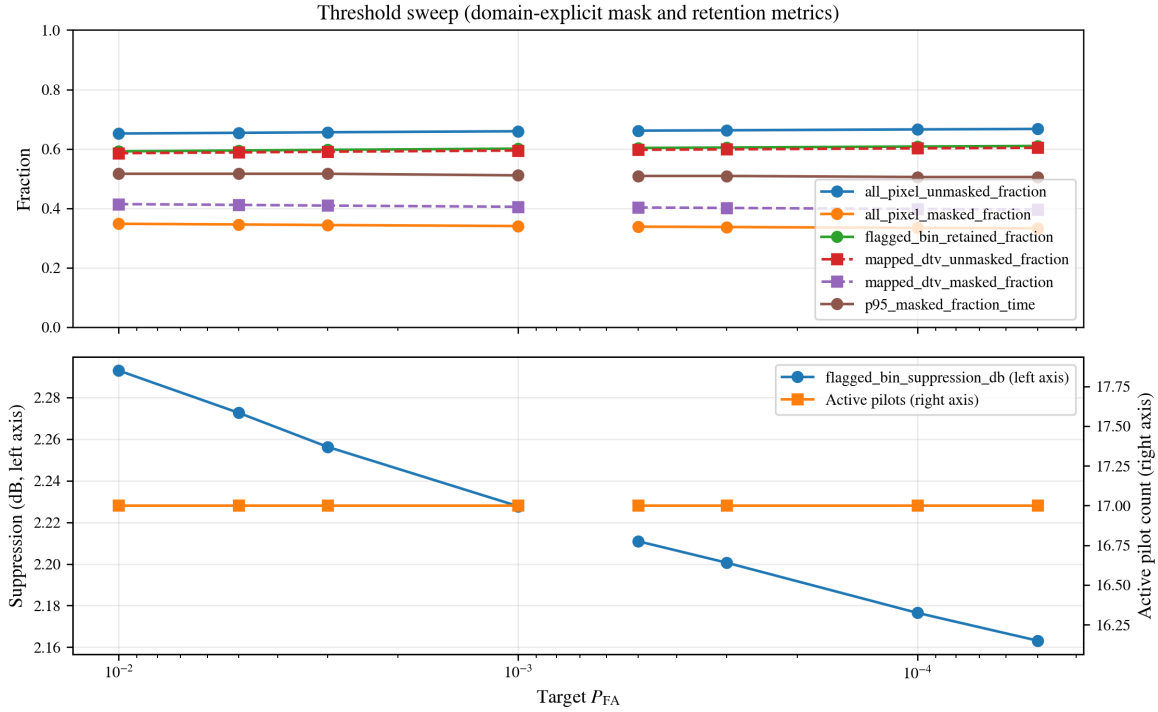


Figure 10: Threshold sweep on the single recorded segment. The upper panel separates all-pixel expanded-mask occupancy, mapped-DTV-bin occupancy, and flagged-bin retained fraction. The lower panel shows suppression over the flagged-bin diagnostic and active-pilot count. These diagnostics are useful for engineering threshold selection but are not archive-scale validation.

9.3 Shifted-grid field-background diagnostic

As a compact real-data background check, we repeated the detector evaluation on the same recorded segment using an ATSC pilot grid shifted by +1.5 MHz. The shifted run uses shifted detector weights, not relabeled nominal scores. Non-nominal shifted guard rows are excluded in the same manner as the nominal publication subset. This diagnostic is not an expanded DTV-mask product, because the shifted grid does not correspond to physical ATSC pilot locations; it is a test of nearby field-background structure relative to the ideal F-null threshold.

Table 3: Shifted-grid field-background diagnostic on the same 0.404 s recorded segment. The detector grid is shifted by +1.5 MHz and evaluated with the same $K = 128$, nominal guard=4, and $P_{\text{FA}} = 10^{-3}$ threshold as the main recorded-data demonstration. The diagnostic is not consistent with the ideal analytic null; it shows that nearby non-pilot locations in this short segment contain substantial real spectral structure.

Quantity	Value
Shift applied to pilot grid	+1.5 MHz
Available shifted rows	16
Strict guard=4 shifted rows used	15
Excluded non-nominal shifted row	549.809 MHz
Time blocks	4930
Tests used	73,950
Threshold	$T = 1.003712$
Analytic target P_{FA}	10^{-3}
Expected exceedances at analytic target	73.95
Observed exceedances	30,023
Empirical exceedance rate	0.406
95% normal-approximation interval	0.402–0.410
Empirical 99.9% shifted-grid quantile	$T = 2.235$

The shifted-grid result is deliberately interpreted as a field-background diagnostic rather than a passing null validation. The empirical exceedance rate is far larger than 10^{-3} , indicating that the short recorded segment contains local spectral structure at nearby non-ATSC shifted positions. Therefore the recorded-data mask panels should be read as demonstration products using the synthetic calibrated threshold, not as an empirical field-null false-alarm validation. A cleaner empty-band or long-duration real-data null study remains necessary for a telescope-scale threshold claim.

An averaged-spectrum sanity check is included in Appendix A. It is kept out of the main results because the short segment is not long enough to support a spectrum-level suppression claim.

10 Scope limitations and required validation

The results support the detector as a calibrated prototype, but several analyses are required before telescope deployment or a stronger instrumentation claim. Table 4 separates what is demonstrated here from what remains to be validated.

Table 4: Current evidence and required follow-on work under the scoped prototype framing.

Topic	Demonstrated in this paper	Next validation step
Null calibration	Effective F null validated with 300,000 simulated null trials; shifted-grid field-background diagnostic reported for one recorded segment	Empty-band and long-duration field-null tests; QQ/tail checks on telescope data
Sensitivity	Injection curves under the stated data-shelf SNR convention	Independent reproduction with archived normalization metadata and matched baselines
Reference geometry	Nominal ± 4 -bin references in detector calibration; non-nominal edge-channel rows excluded from the recorded-data publication subset	Guard sweep over wider spacings, e.g., $g = 6, 8, 10, 12, 16$, comparing leakage, field-background exceedances, and strict-row availability
Baseline comparison	Reserved for validation study	Matched finite-sample local-energy detector with same trials, P_{FA} , and SNR convention
Recorded data	0.404 s single-segment output demonstration with 17 nominal guard=4 pilot products, masks, histograms, threshold sweeps, and shifted-grid diagnostic	Multi-hour or multi-window validation across RFI regimes
Mask burden	Single-segment threshold sweep and false-alarm accounting	Science-loss threshold selection and cross-stream aggregation policy
GPU implementation	Fixed-point CUDA prototype benchmarked kernel-only	Optimized real-time implementation including transfer and integration overheads
Science impact	Detector-side products only	BAO/noise degradation mapping, e.g., through Fisher forecasting or end-to-end pipeline tests

The matched energy-detector baseline should use the same fine-bin representation, same accumulation length, same P_{FA} calibration, same injection grid, and same recorded segments. The point of that comparison is not only sensitivity. A pilot-tone detector also tests an ATSC-specific feature, whereas a channel-energy detector responds to any broad local excess. This distinction is central to the final novelty argument and should be included in a later full validation paper.

For archive-scale validation, a useful minimum target is several independent windows totaling at least several hours of data and spanning quiet, representative, and high-RFI regimes. The validation should report active pilot count versus time, mask fraction versus frequency and time, empty-band or field-null score distributions, before/after spectra, residual pilot scores, threshold sweeps, and failure cases such as adjacent-channel leakage or reference-row contamination. The shifted-grid diagnostic included here shows that nearby recorded-data structure can strongly exceed the ideal null; a cleaner empty-band or long-duration field-null analysis remains useful for deployment validation. A guard sweep is also on the deployment to-do list, because larger guard offsets may provide additional leakage margin but may also sample less local spectral structure.

11 Discussion

The central technical result is that an ATSC pilot feature can be expressed as a local F-ratio detector with an interpretable CFAR threshold. The simulated null distribution agrees with the effective F model over the sampled tail range, and the score threshold can be reproduced directly from P_{FA} . This is useful for instrumentation because mask generation becomes an engineering choice on a calibrated detector axis rather than a sequence of ad hoc power cuts.

The present evidence is intentionally focused on detector calibration and implementation feasibility. The single recorded segment confirms that the code path produces plausible pilot-score, pilot-distribution, expanded-mask, and shifted-background diagnostic products. The shifted-grid result also shows that nearby non-pilot spectral structure in this field is much broader than the ideal analytic null, so the recorded-data demonstration should not be read as a field-null false-alarm validation. Longer runs are needed to probe stability, empty-band false-alarm behavior, and operational mask burden. Likewise, the sensitivity curves are internally calibrated, while a matched local-energy comparison is reserved for validation work. These scope boundaries separate the detector-calibration contribution from a later deployment-validation paper.

The fixed-point CUDA result is also best interpreted as a feasibility result. Kernel-only throughput of 29.02 GB/s for packed input payload is substantial, and the fixed-point residuals are bounded over the tested range, but the kernel remains slower than the present real-time block-rate requirement. Runtime breakdown indicates that accumulation dominates, suggesting that memory layout, batching, stream fusion, and integration with upstream channelized data products will likely matter more than the F-ratio arithmetic itself. The natural follow-on sequence is therefore a science-threshold paper that maps mask burden to BAO tolerance and a deployment paper that treats long-duration data, matched baselines, guard sweeps, and optimized real-time integration.

12 Conclusions

We presented a calibrated ATSC pilot-tone F-ratio detector and fixed-point GPU prototype for DTV flagging in radio astronomy data products. The detector uses one target matched row and two guard-separated reference rows around expected ATSC pilot locations and is calibrated with an effective F-distributed null model. For the current $K = 128$, nominal guard=4 configuration, the null model agrees with 300,000 simulated null trials at the empirically supported tail depth.

Injection tests show that the detector can recover weak pilot-associated DTV contamination under the stated data-shelf SNR convention; robustness diagnostics quantify frequency-offset, gain/noise, and reference-leakage effects. The CUDA prototype demonstrates a fixed-point implementation path and identifies the remaining throughput shortfall. A short CHIME baseband-derived segment demonstrates pilot-level detection, expanded DTV-channel mask generation, threshold sweeps, per-pilot pilot-power distributions, and a shifted-grid field-background diagnostic. A recomputed +1.5 MHz shifted-grid diagnostic shows substantial field-background structure in that segment, so the recorded-data panels are treated as demonstration products rather than field-null false-alarm validation.

This prototype paper establishes the detector, calibration, plotting conventions, mask rule, and prototype implementation. The next required steps are a matched energy-detector baseline, clean empty-band or long-duration field-null validation, recorded-data testing across multiple RFI regimes, a guard-spacing sweep for deployment margin, and threshold selection tied to a science-loss budget.

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Code, data, and reproducibility

The manuscript source package includes the detector configuration, regenerated figures, tabulated within-bin loss curve, JSON result summaries, histogram products, threshold-sweep cache, and detector weight table used for the prototype calculations. The result package used for this revision was generated as run 20260509T022751Z with git hash `bdf2814da44018cba2d3183bc4d86e85a54562b9`. The detector configuration is $K = 128$, $M = 1,048,576$, three weight vectors (target, lower reference, upper reference), nominal reference offsets $\{-4, +4\}$, $\nu_1 = 2,097,152$, $\nu_2 = 4,194,304$, and default $P_{\text{FA}} = 10^{-3}$ threshold $T = 1.003712$; the value of M corresponds to $2^{16}/128$ subblocks over 1024 antennas and two polarizations. The real-data figures use a 0.404 s CHIME segment and a strict guard=4 publication subset with 17 nominal pilot rows; four expected pilot channels are unavailable in the cache and two non-nominal edge-channel rows are explicitly excluded. The package also includes a +1.5 MHz shifted-grid field-background diagnostic with 73,950 strict-guard tests and empirical exceedance rate 0.406. The included preflight summary reports that all required generated-figure checks pass. For a formal public release, the local source package should be deposited with a repository DOI, checksum, and formal data-access statement.

A Averaged-spectrum diagnostic for the single recorded segment

Figure 11 gives an averaged-spectrum sanity check using the same short segment as Fig. 8. The top panel compares the median spectrum before masking with the median spectrum after retaining unmasked samples. The bottom panel shows retained fraction versus frequency. In this short example the before/after spectra remain similar, which is expected because the segment is brief and

much of the strongest structure is persistent. Long-duration validation should repeat this calculation across many windows so that spectrum-level suppression and possible mask-induced selection effects can be quantified statistically.

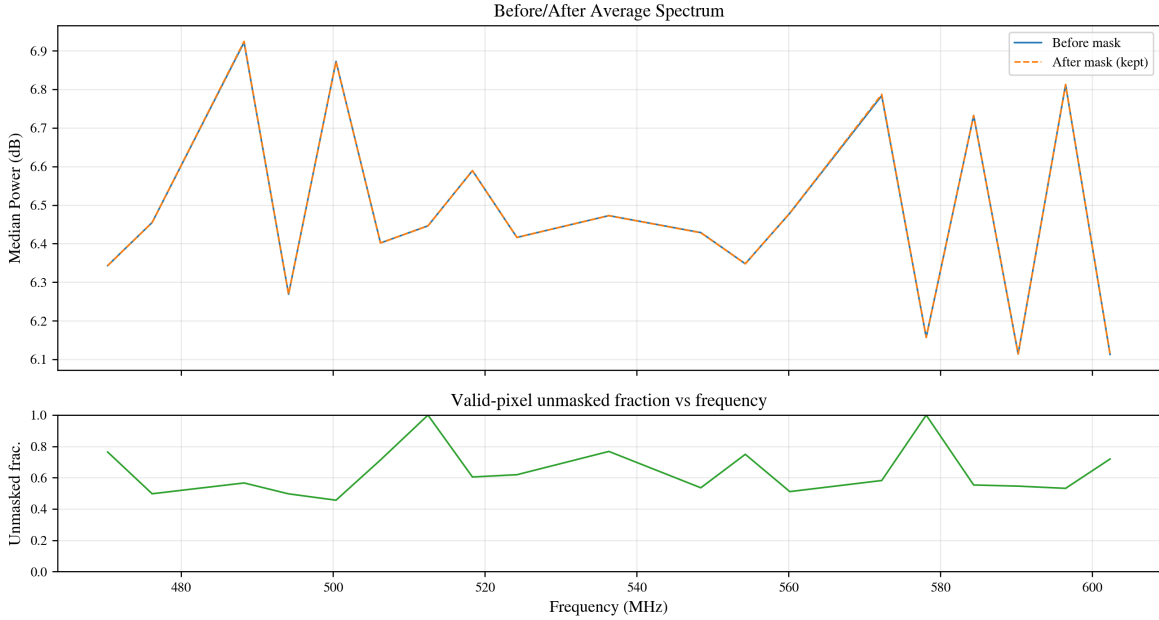


Figure 11: Averaged-spectrum effect of applying the expanded DTV mask to the same recorded segment used in Fig. 8. The upper panel compares the median spectrum before masking and after retaining unmasked samples. The lower panel shows retained fraction versus frequency. The short segment gives a useful sanity check but does not constitute a spectral-flattening performance claim.

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